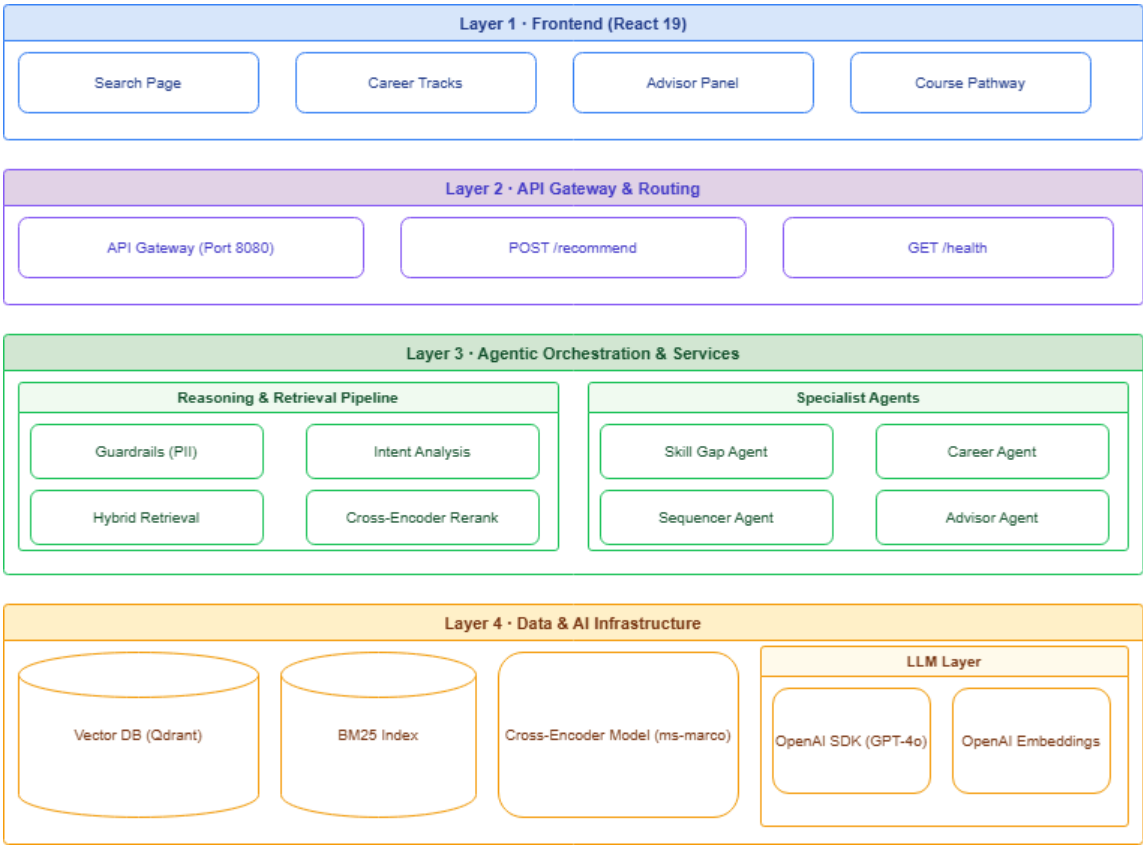
Intelligent University Course Finder

An AI-powered discovery system that helps students navigate university courses effectively. By combining Hybrid Retrieval (Vector + Keyword) with a pipeline of specialized AI Agents, the system understands messy human intents, identifies skill gaps, aligns courses to career tracks, and generates a personalized, difficulty-staged learning path.

🌟 Key Features

- **Agentic Pipeline (Parallel Processing)**: Built with the OpenAI Agents SDK. Runs `SkillGapAgent` and `CareerAgent` in parallel, cutting logical latency in half.
- **Hybrid Retrieval Framework w/ Reranking**: Fuses Qdrant Dense Vector Search (semantic) with BM25 Keyword Search (exact token match) using Reciprocal Rank Fusion (RRF), capped off with **Cross-Encoder Reranking** (`ms-marco-MiniLM-L-12-v2`) to intelligently sort final candidates.
- **Parent/Child Domain Chunking**: Courses are broken down into Identity, Skills, and Description "child" query variants, while the entire rich "parent" course object is returned to the agent without noisy embedding pollution.
- **Agentic Course Sequencer**: An LLM-based `SequencerAgent` that reasons about conceptual dependencies and career goals to design themed learning milestones (e.g., "Python Foundations" -> "Data Science Core").
- **Pre-Emptive Security Guardrails**:
  - **Presidio Analyzer / Anonymizer**: Detects and redacts PII elements client side *before* any prompt leaves the local machine.
  - **LangChain Semantic Intent Validation**: Rejects off-topic, toxic, or prompt-injection queries pre-emptively on the gateway.
- **Modern Microservice Architecture**:
  - FastAPI **API Gateway** acting as an intermediate decoupling surface processing timeouts.
  - Dedicated **FastAPI Microservice** orchestrating AI Logic and Database calls.
  - **React + Vite** frontend implementing a clean, modular minimalist light theme.

The system is organized into a layered, block-wise structure.



You can view the full detailed breakdown in **Artifacts/Architecture Diagram.png**.

System Layers

1. **Frontend (React)**: Component-driven interface (Search, Career Card, Advisor Summary, Course Path).
2. **API Gateway (FastAPI)**: Single proxy endpoint (Port 8080) for frontend isolation and traffic logging.
3. **Core API (FastAPI)**: Central Microservice (Port 8000) orchestrating agent workflows.
4. **Guardrails (Presidio & LangChain)**: Pre-processing queries to remove PII and validate learning intent.
5. **Agent Pipeline (OpenAI SDK)**:
   - *IntentAgent*: Normalizes raw queries into structured `{topic, level, keywords}` filters.
   - *SkillGapAgent*: Identifies missing prerequisites.
   - *CareerAgent*: Aligns courses to job tracks.
   - *SequencerAgent*: Designs intelligent, themed learning paths.
   - *LearningAdvisorAgent*: Summarizes and structures actionable human-friendly tips.
6. **Retrieval**: Qdrant Vector DB & BM25 Pickle Index + Cross-Encoder Reranking.

> 🗂️ *For a file-by-file breakdown, refer to the **Artifacts/CODE_STRUCTURE.md** and the design rationale in **Artifacts/design_decisions.md**.*

🚀 Getting Started

Prerequisites

- Python 3.10+
- Node.js (v18+) & NPM

- OpenAI API Key

1. Installation

Clone the repo, install python packages, and install UI packages:

```
# Backend python dependencies
pip install -r requirements.txt

# Download Spacy model for Presidio PII Detection
python -m spacy download en_core_web_lg

# Install frontend dependencies
cd frontend/careerguidefrontend
npm install
cd ../..
```

2. Configuration

Create a `.env` file in the root directory:

```
OPENAI_API_KEY=sk-...
LLM_MODEL=gpt-4o
QDRANT_PATH=storage/qdrant_db
BM25_INDEX_PATH=storage/bm25_index.pkl
FRONTEND_URL=http://localhost:5173
```

3. Data Ingestion

You must initialize the Vector & Keyword databases before querying.

```
python ingestion/ingest.py
```

This will parse `data/coursera_courses.csv`, chunk documents natively into parent/child formats, embed them with OpenAI text-embedding-3-small, upsert to `storage/qdrant_db`, and pickle the `storage/bm25_index.pkl` keyword store.

💻 Running the Application

To start the full environment, you will need **3 terminal tabs**.

Terminal 1: Start the Backend Microservice (Port 8000)

```
uvicorn api.main:app --port 8000 --reload
```

Terminal 2: Start the API Gateway (Port 8080)

```
uvicorn api_gateway.gateway:app --port 8080 --reload
```

Terminal 3: Start the UI / Frontend (Port 5173)

```
cd frontend/careerguidefrontend
npm run dev
```

Finally, open `http://localhost:5173` in your browser.

🧪 Testing

Standalone test scripts are located in the `/test` directory. Use them to debug specific nodes in the graph in isolation.

- **Check Qdrant DB points/connections**: `python test/check_qdrant.py`
- **Test Hybrid Retriever & Filters**: `python test/test_retriver.py`
- **Test Guardrails (PII & Semantic Intent)**: `python test/test_guardrails.py`
- **Test IntentAgent (Structured Extraction)**: `python test/test_intent_agent.py`

📝 Logging

The application utilizes centralized structured logging logic via `logger.py`. Logs are separated per-module in the `/logs` directory (e.g., `logs/guardrails.log`, `logs/intent_agent.log`) enabling granular debugging without polluting output.